# Neuroprosthetics
# Exercise Six

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## 1 Task One: Potential Field

As done in the last exercise, the goal of this exercise is to simulate action potential propagation along the axon of a neuron. In the last exercise, it was assumed that a stimulating current acts directly upon our axon. In reality, stimulating currents do not do this, but rather induce potentials in the action, which then cause the stimulation and setting off of action potentials. To mathematically express this, we must take a look again at the equivalent circuit of an axon to derive the cable equation.

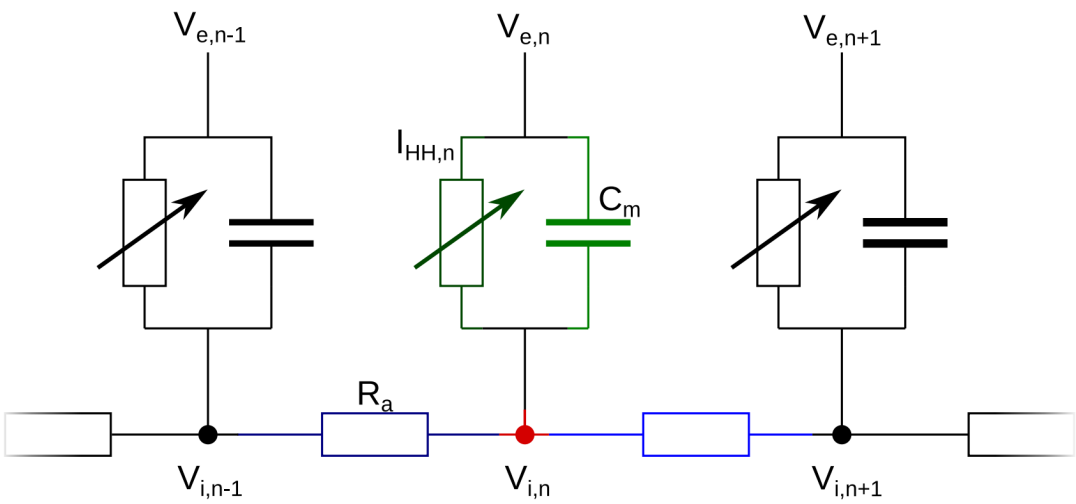


Figure 1: Equivalent circuit for an axon denoting the cable equation dynamics. Figure taken from the lecture slides.

We know that potentials from neighbouring compartments influence each compartment in our axon. As seen in fig. 1, we do not only take into consideration $V_{i,n-1}$ and $V_{i,n+1}$, but we must also consider the external potentials $V_{e,n-1}$, $V_{e,n}$, and $V_{i,n+1}$. In the

past exercise, we omitted them and introduced a stimulus current. In this exercise, we will use a stimulus current to calculate our external potentials $V_e$.

To first derive the cable equation, let us recall our result from the past exercise.

$$\frac{dV_{m,n}}{dt} = \frac{1}{C_m}(-I_{HH,n} + I_{stimulus}) + \frac{1}{C_m} \cdot \frac{V_{i,n-1} - 2V_{i,n} + V_{i,n+1}}{R_a}. \tag{1}$$

Adapting this equation to model properly our circuit, we must remove the stimulus current and introduce our external potentials. The resulting equation looks as follows:

$$\frac{dV_{m,n}}{dt} = \frac{1}{C_m}(-I_{HH,n}) + \frac{1}{C_m} \cdot (\frac{V_{i,n-1} - 2V_{i,n} + V_{i,n+1}}{R_a} + \frac{V_{e,n-1} - 2V_{e,n} + V_{e,n+1}}{R_a}). \tag{2}$$

We can directly recognise that the $V_e$ terms have the exact same coefficients as the $V_i$ terms. In this sense, we can also write our equation for all compartments using the connection matrix $\mathbf{C}$ for both internal and external potentials.

$$\frac{d}{dt}\vec{V}_m = \frac{1}{C_m}(-I_{HH,n}) + \frac{1}{C_m R_a} \cdot \mathbf{C} \cdot (\vec{V}_i + \vec{V}_e) \tag{3}$$

As a reminder, the connection matrix $\mathbf{C}$ is the following:

$$\mathbf{C} = \begin{pmatrix} -1 & 1 & & & & & 0 \\ 1 & -2 & 1 & & & & \\ & \ddots & \ddots & \ddots & & & \\ & & 1 & -2 & 1 & & \\ & & & \ddots & \ddots & \ddots & \\ & & & & 1 & -2 & 1 \\ 0 & & & & & 1 & -1 \end{pmatrix}$$

The term for the external potentials in eq. 2 approximates the second derivative of the potentials themselves. This function is called the activation function, as it is the stimulant that decides if action potentials are triggered in the axon.

## 1.1 Activation Function

To know how our activation function, and consequently our action potentials, behave, we must first be able to calculate the external potentials that are induced by stimulation currents. The formula

$$\Phi = \frac{\rho}{4\pi} \cdot \frac{I}{r} \tag{4}$$

lets us calculate the induced potential in an axon, when a current source is placed some distance $r$ away from the axon. $I$ is then the current flowing from our source (in our case, an electrode), while $\rho$ is the resistivity of the medium between electrode and axon.

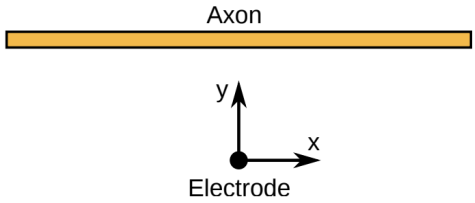


Figure 2: Electrode-Axon configuration. Figure taken from the exercise handout.

From our configuration, which is to be seen in fig. 2, we can then further write our potential equation as

$$\Phi = \frac{\rho}{4\pi} \cdot \frac{I}{\sqrt{x^2 + y^2}}. \tag{5}$$

To arrive at the activation function, we need to derive our potential twice. The first derivative of potential equals the negative electric field produced:

$$\frac{d\Phi}{dt} = -\vec{E} \tag{6}$$

In our case, as seen in fig. 2, we have a constant distance in the y direction between our axon and the electrode. However, as the axon is placed along the x direction, this means that the distance in this direction between electrode and the individual compartments changes. Compartments at the middle of the axon will have a bigger potential than those at the tips. As we can see that the distance $y_r$ is constant, our derivative is only dependent on x. Deriving by $dx$ results in the following equation for the electric field induced on the axon:

$$\frac{d\Phi}{dx} = -\frac{\rho I}{4\pi} \cdot \frac{x}{(\sqrt{x^2 + y^2})^3} \tag{7}$$

which means that the electric field induced is

$$\vec{E} = \frac{\rho I}{4\pi} \cdot \frac{x}{(\sqrt{x^2 + y^2})^3}. \tag{8}$$

Furthermore, we know that the activation function is the **positive** second derivative of the potential (as opposed to the electric field being the **negative** first derivative). We then derive eq. 7 once more with respect to x.

$$\frac{d^2\phi}{dx^2} = \frac{\rho I}{4\pi} \cdot (-\frac{3x^2}{(\sqrt{x^2 + y^2})^5} + \frac{1}{(\sqrt{x^2 + y^2})^3}). \tag{9}$$

With these equations, we can then calculate and plot the graphs of the external potentials, the electric fields they induce, and the consequent activation function.

For this, we are using following parameters: a medium resistivity $\rho_{medium} = 1m$, a first stimulus current $I_1 = 1mA$, a second stimulus current $I_2 = -1mA$, a distance in the y direction $y_r = 10\mu m$, and an axon compartment length $l_{comp} = 0.1\mu m$. The axon itself possesses a total axon length $l_{axon} = 300\mu m$. This means, we will have to check every 0.1 $\mu m$ how each of our functions change.

The potentials produced along the axon can be seen in the figures below for both $I_1$ and $I_2$.

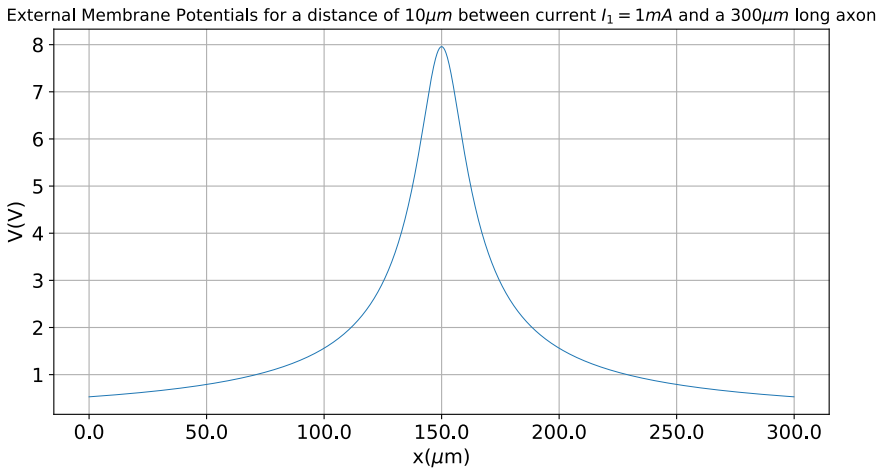


Figure 3: External Potentials over the length of an axon using the first stimulus current

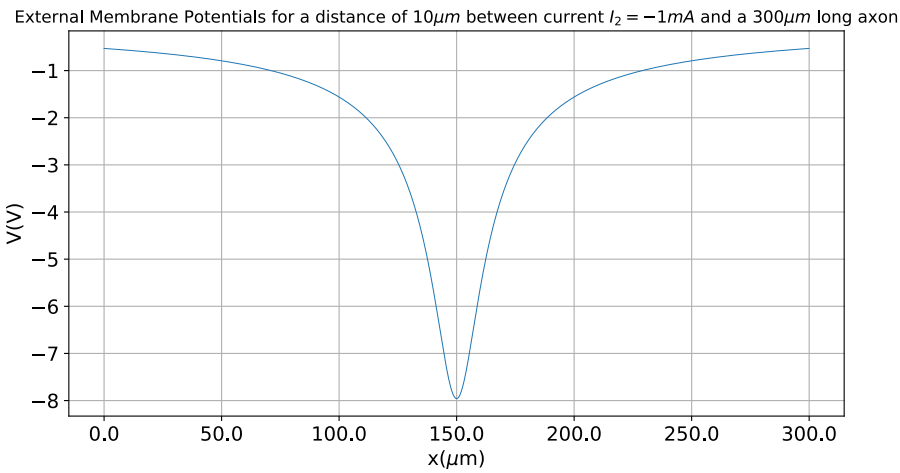


Figure 4: External Potentials over the length of an axon using the second stimulus current

It can be easily seen that potentials peak where the electrode is closest to the axon, while compartments that find themselves at the ends have much lower external potentials. This means, compartments closer to the electrode will have a higher chance of firing action potentials, as we will see soon enough. It is important to note the sign of the functions for each potential, as it will have an influence later on as well. The electric fields for both configurations follow.

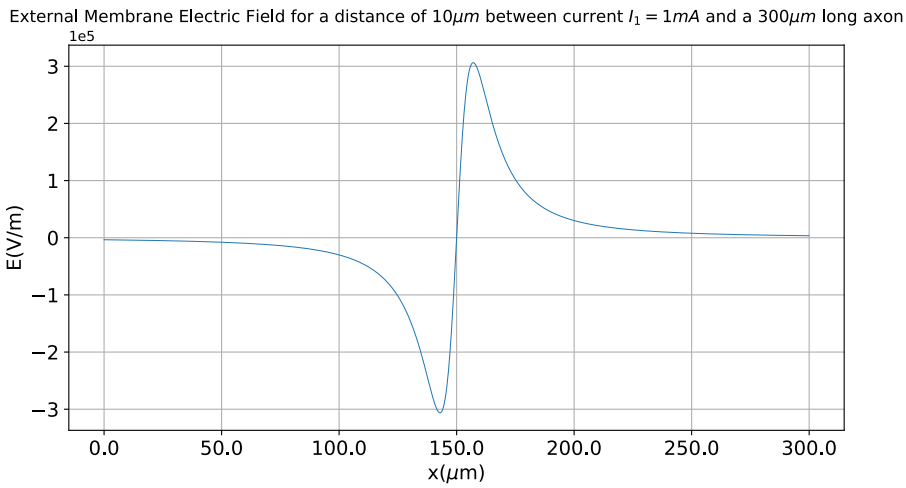


Figure 5: Electric Field values over the length of an axon using the first stimulus current

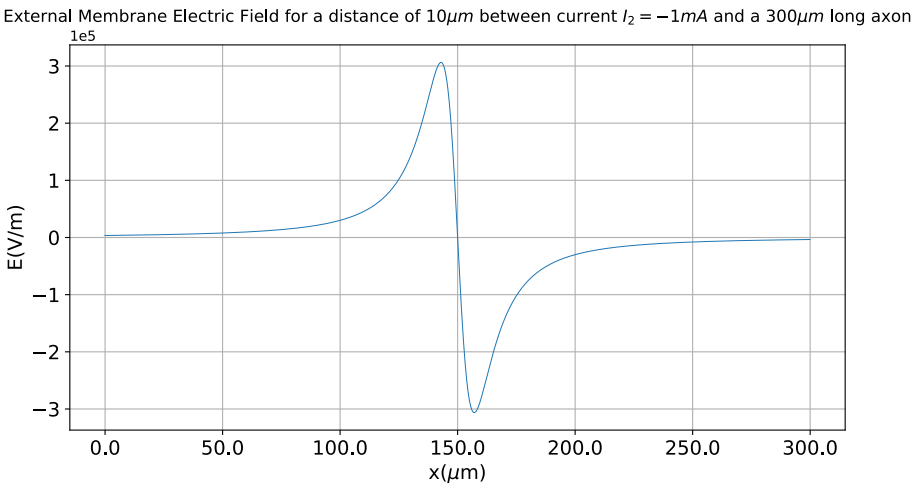


Figure 6: Electric Field values over the length of an axon using the second stimulus current

Lastly, we can plot the activation function values for each of the stimulus currents $I_1$ and $I_2$.

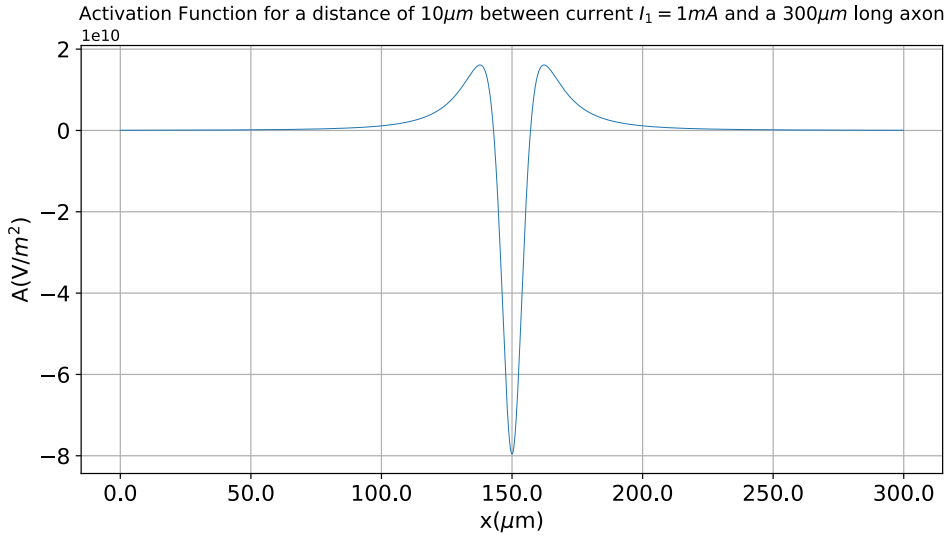


Figure 7: Activation Function values over the length of an axon using the first stimulus current

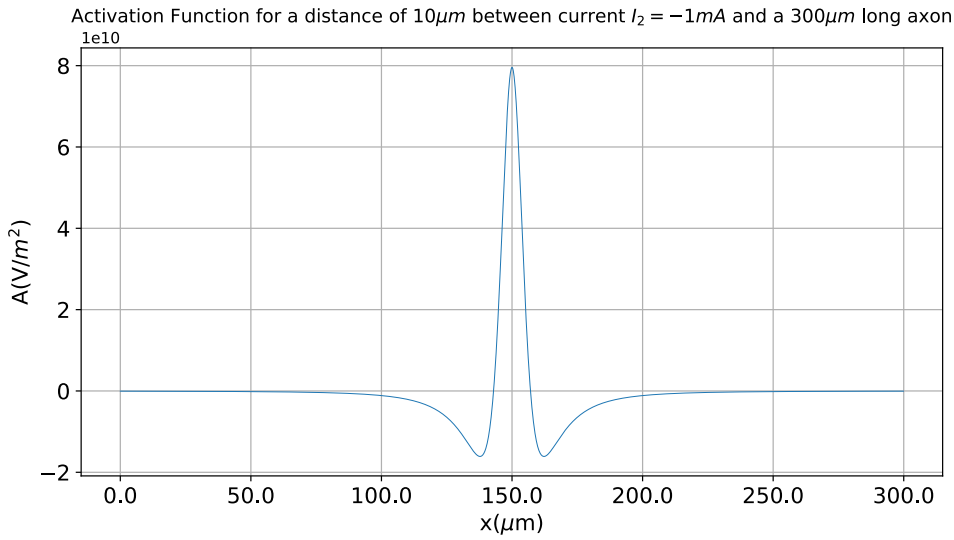


Figure 8: Activation Function values over the length of an axon using the second stimulus current

From fig. 7, which comes from the use of a positive $I_1 = 1mA$ stimulus current, we can see that the activation values of the axon peak to the sides of the middle with relatively low values, while the middle of the axon experiences what we could call a notch, making it extremely difficult for any action potentials to happen in the middle compartments. The relatively low peak values mean that a higher current would be needed to trigger an action potential.

Looking at fig. 8, which used the negative $I_2 = -1mA$ stimulus current, we see the same function as in fig. 7 but mirrored on the x-axis. This means, that now the middle of the axon experiences very high activation values, even with the same magnitude of current, just by inverting its direction. As we will see in later experiments in this report, this influences how easy it becomes to trigger action potentials when using a negative stimulus current.

## 2 Task Two: Neuron Model

Having derived the equation 3 using only external potentials $V_e$ as stimulants, we must now adapt it to the Implicit Euler numerical solver method for our simulation. Recall the Implicit Euler method:

$$V(t + \Delta t) = V(t) + \Delta t \cdot f(t + \Delta t) \tag{10}$$

When we apply this to eq. 3, we obtain

$$\vec{V_m}(t + \Delta t) = \vec{V_m}(t) + \frac{\Delta t}{C_m}(-\vec{I}_{HH}(t + \Delta t)) + \frac{\Delta t}{C_m R_a}\mathbf{C}(\vec{V}_m(t + \Delta t) + \vec{V}_e(t + \Delta t))) \tag{11}$$

which we can then rewrite as

$$(I - \frac{\Delta t}{C_m R_a}\mathbf{C}) \cdot \vec{V}_m(t + \Delta t) = \vec{V_m}(t) + \frac{\Delta t}{C_m}(-\vec{I}_{HH}(t + \Delta t)) + \frac{\Delta t}{C_m R_a}\mathbf{C}\vec{V_e}(t + \Delta t) \tag{12}$$

fully replacing the earlier stimulus current input with the more accurate external potentials. The rest of the code and the simulation stay the same.

For our experiments, the same parameters were used as in the last exercise, except for the following changes in axon parameters: a new axoplasmic resistivity $\rho_{axon} = 7 \cdot 10^{-3} m$, a new axonal radius $r_{axon} = 1\mu m$, and a new compartment length $l_{comp} = 3\mu m$. These parameters mostly affect the final axonal resistance $R_a$, which is calculated in the same manner as in the last exercise, and the calculation of the external potentials $V_e$, as we now use a resolution of $3\mu m$, as opposed to the $0.1\mu m$ used in task one of this exercise.

Each experiment runs for a total time of $t_{total} = 30ms$, with a $\Delta t = 25\mu s$ time step. All stimulation pulses start at $t_{start} = 5ms$. We used six different input currents to analyse the behaviour of the axon and the influence of the activation function onto the triggered action potentials. The first current $I_1 = -0.05mA$ is a mono-phasic current with a phase duration of $1ms$. This yields the following graph:

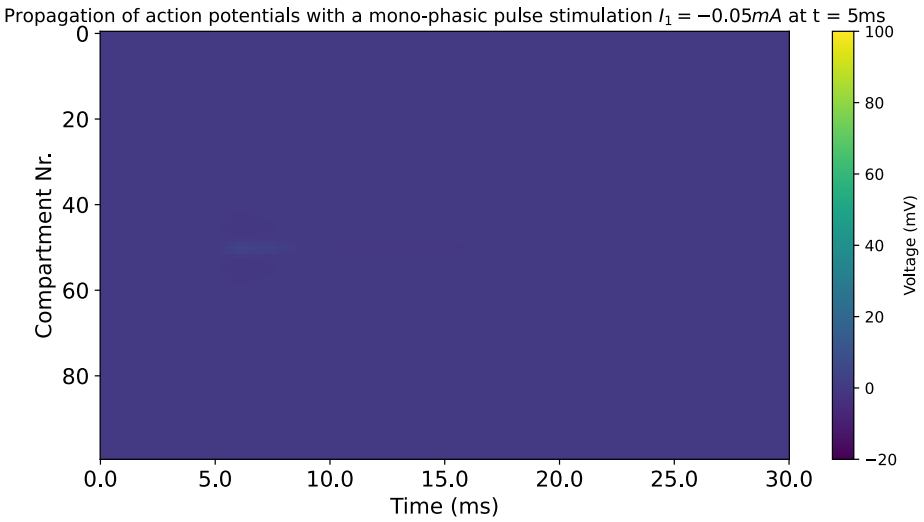


Figure 9: Action Potentials along a neuron's axon for a mono-phasic pulse stimulation $I_1 = -0.05mA$

Several conclusions can be drawn from fig. 9. Using a relatively low current, we can see a change in voltage in the middle compartment around $t = 5ms$. The pulse is neither strong nor long enough to elicit action potentials, but even with its short duration and low magnitude, we can see its influence on the middle compartment. This aligns well with fig. 8, as our stimulus current in this case is negative, and we see the middle compartments being easily influenced thanks to the very high values of the activation function in the middle of the axon.

The second stimulus pulse $I_2 = -0.1mA$ is a mono-phasic pulse as well. The results of the experiment are below.

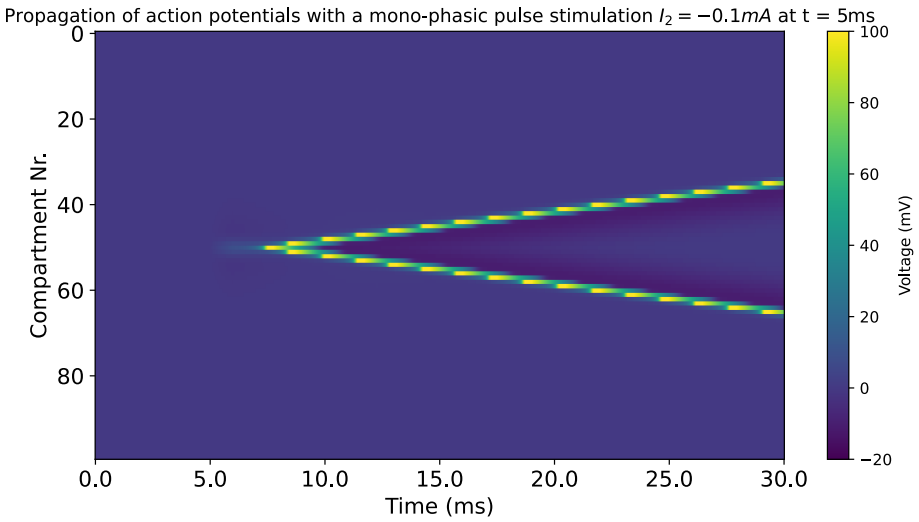


Figure 10: Action Potentials along a neuron's axon for a mono-phasic pulse stimulation $I_2 = -0.1mA$

Fig. 10 shows us what happens if we simply double the magnitude of the first current. Consistent with fig. 8 and the first pulse, we see that slightly increasing the current's magnitude elicits an action potential in the middle that travels in both directions along the axon. Just as with the first current, the potential happens where the activation function shows the highest values, which is the middle compartments of the axon. The behaviour of the axon is consistent and expected.

For the third experiment, a bi-phasic stimulus current $I_3 = \mp 0.1mA$ is used. A bi-phasic pulse is one that has the same magnitude over both phases that comprise it, albeit with a flipped sign for each phase. As stated before, each phase lasts $1ms$, meaning that this pulse has a total duration of $2ms$ for its two phases. The starting phase is negative, followed by the positive phase. The results are the following:

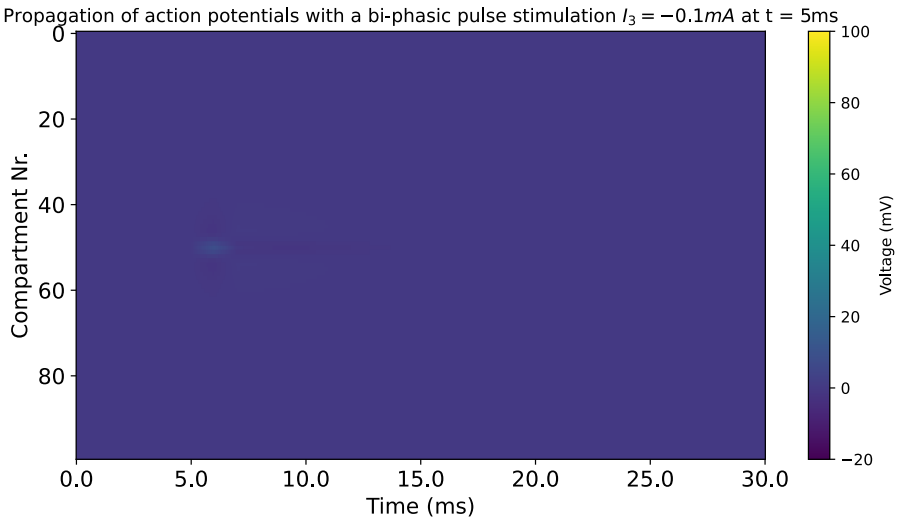


Figure 11: Action Potentials along a neuron's axon for a bi-phasic pulse stimulation $I_3 = \mp 0.1mA$

The case of fig. 11 is quite interesting: from fig. 7 showing us a positive current's activation function, we can see that the middle of the axon would experience hyperpolarisation while any activation would happen slightly at the sides of the middle; and fig.8, for a negative stimulation current, shows us the opposite behaviour: activation in the middle with the sides of the middle probably going into a refractory period. We can see this in fig. 11 directly around the $5ms$ mark: a higher voltage in the middle with the neighbouring compartments showing a darker tone, a sign for lower potentials. As the second phase starts, which has a positive magnitude, the middle compartments turn a darker colour, signifying a lower voltage and possibly slight hyperpolarisation. The neighbouring compartments, which previously were toned in a darker shade, experience a slight increase in voltage, consistent with the activation function for a positive current. What is most interesting, is that even if the first phase is the exact same current as in the last experiment, no action potentials happen. This can be explained by the second phase cancelling out any voltage accumulated from the first phase before any APs can occur.

For the fourth current, the same bi-phasic pulse as in the third experiment was used, although with a higher magnitude $I_4 = \mp 0.15mA$.

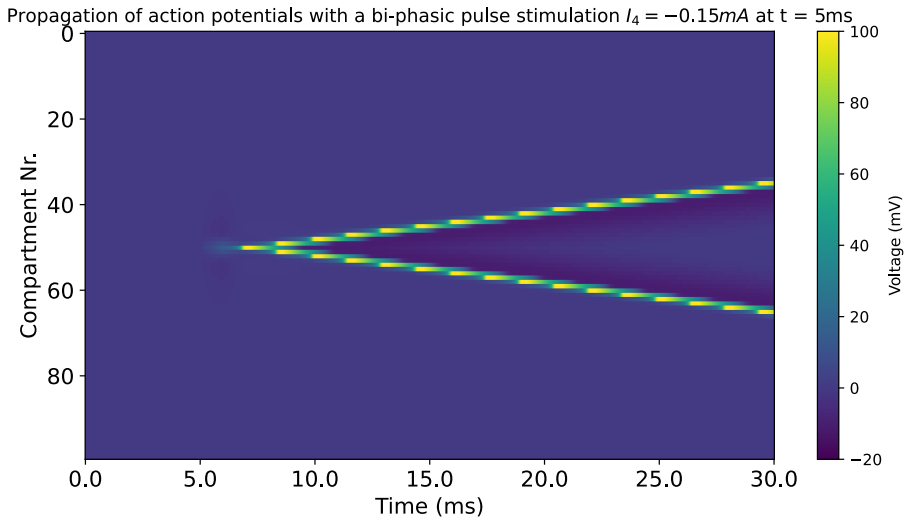


Figure 12: Action Potentials along a neuron's axon for a bi-phasic pulse stimulation $I_4 = \mp 0.15mA$

For fig. 12, we see what would happen if the second phase of the third current was not able to cancel out the first phase. The increased magnitude in current means that the first phase of our pulse manages to trigger an action potential before the second phase is able to lower the voltage values in the middle compartments, which is what happened with the third stimulus pulse. The neighbouring compartments around the action potential still increase their voltages thanks to the behaviour of the activation function for a positive stimulus current.

The fifth stimulus pulse is again a mono-phasic current $I_5 = 0.2mA$. The results are seen below.

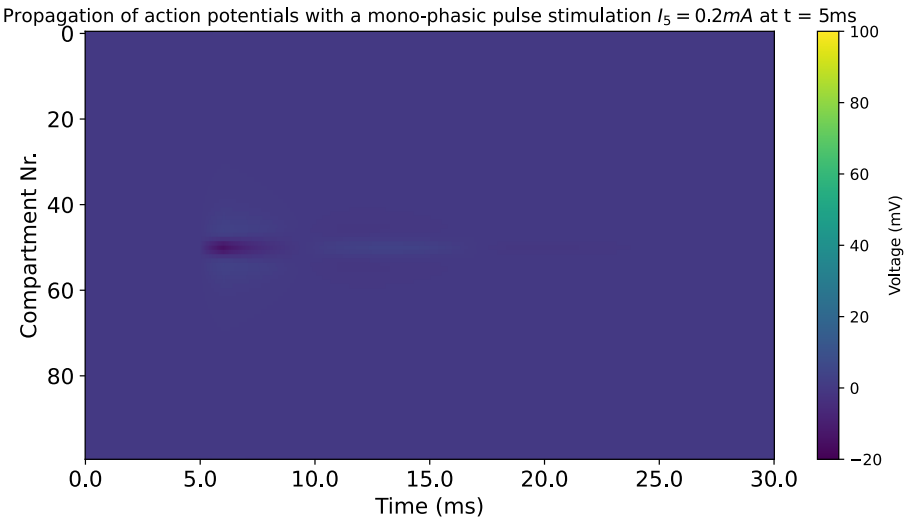


Figure 13: Action Potentials along a neuron's axon for a bi-phasic pulse stimulation $I_5 = 0.2mA$

In fig. 13 we see interesting behaviours. The first point is the consistency with fig. 7: the middle compartments go into hyperpolarisation while the neighbouring ones increase in voltage, as explained by the respective activation function. After a short period of time, it seems that the middle compartments recover and also experience a rise in their potentials. This could happen thanks to the higher voltages from the neighbouring compartments. As we know from previous exercises, the potential in one compartment is influenced by its neighbouring compartments, and as the middle compartments leave their hyperpolarised state, it is possible that their neighbours help in raising their voltages above resting potential for a short period of time without firing, before going again into a refractory period. One last thing to point out is how a positive current that has double the magnitude of current $I_2$ is not yet capable of eliciting any action potentials. This is consistent with the activation functions: negative currents make the triggering of potentials much easier with lower currents, while positive currents need much higher currents.

The sixth and final stimulus current is a mono-phasic current, with the highest magnitude of any of our currents $I_6 = 0.4mA$. The resulting action potentials can be seen below.

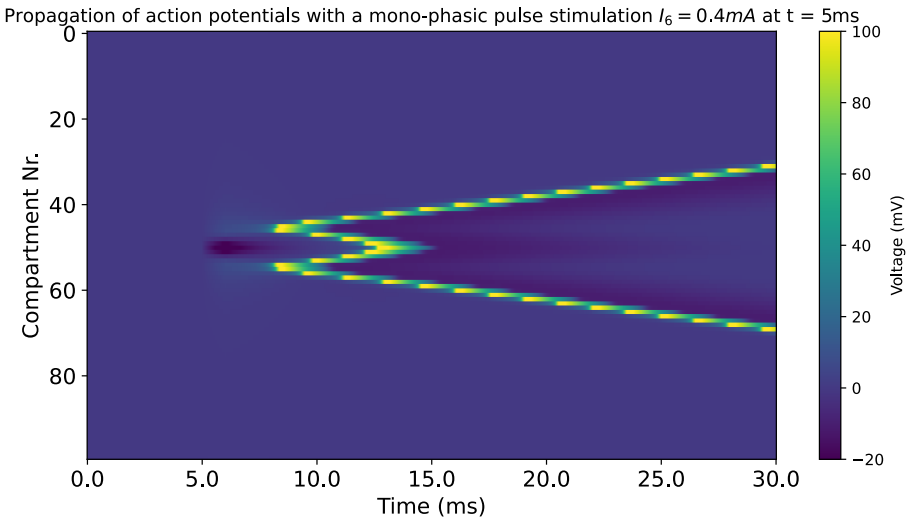


Figure 14: Action Potentials along a neuron's axon for a bi-phasic pulse stimulation $I_6 = 0.4mA$

In this graph we can see that two action potentials happen consistent with the positive current activation function seen in fig. 7: the middle compartments go into hyperpolarisation, while their neighbouring compartments experience action potentials travelling in both directions along the axon. These APs meet once in the middle, cancelling each other out due to the refractory periods of their predecessing compartments, but above and below the middle they extend further along the axon. This behaviour is consistent with the rest of the graphs and explanations, including the fact that a positive sitmulus current will need to be much higher to elicit an action potential in comparison to a negative stimulus current.